

Python Fundamentals for Security

Why Python

Python is the language of security tooling: Metasploit modules, Scapy, Volatility2 plugins, custom exploits, and parsing scripts. Readable, batteries-included, cross-platform.

Essential Types

```
name = "alice"          # str
count = 3                # int
ratio = 0.93             # float
present = True           # bool
items = [1, 2, 3]         # list (ordered, mutable)
record = {"ip": "10.0.0.1", "port": 22} # dict
unique = {1, 2, 2, 3}     # set (no duplicates)
```

Conditionals & Loops

```
for ip in ["10.0.0.1", "10.0.0.2"]:
    if ip.startswith("10.0"):
        print("internal:", ip)
    else:
        print("external:", ip)

while True:
    line = input("data> ")
    if line == "exit":
        break
```

Files

```
# Read a log line by line
with open("access.log", "r") as f:
    for line in f:
        parts = line.split()
```

```
ip = parts[0]
if ip == "192.168.1.50":
    print("SUSPECT:", line.strip())
```

List Comprehensions (Security Gold)

```
ips = [l.split()[0] for l in open("access.log") if "curl" in l]
unique_ips = set(ips)
print(len(unique_ips), "unique IPs used curl")
```

The "parse-everything" habit

Security = parsing. You'll parse logs, packets, configs, hashes, JSON, CSV constantly. Python's `json`, `csv`, `re` (regex) modules make this fast.

Golden rule: understand the data before you analyze it. Run `print(type(x))`, `print(len(x))`, and peek at the first few items.

[← Module start](#)[Web Programming Basics for Sec...](#)